$$\sum_{i=a}^{N} 1 = (N+1) - a$$

$$\sum_{i=1}^{N} i = \frac{N(N+1)}{2}$$

$$\sum_{i=1}^{N} i^2 = \frac{N(N+1)(2N+1)}{6}$$

$$\sum_{i=1}^{N} i^k \approx \frac{N^{k+1}}{|k+1|}$$

$$\sum_{i=n_0}^{N} f(i) = \sum_{i=1}^{N} f(i) - \sum_{i=1}^{n_0-1} f(i), \quad n_0 > 1$$